Discontinuous Galerkin For Sequences of Earthquakes and Aseismic Slip MFEM Implementation

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1 Quasi-Dynamic Antiplane Shear Problem

1.1 Strong Form

The governing equations in physical domain Ω with its boundary $\partial\Omega$. The boundary $\partial\Omega$ is partitioned into Dirichlet boundary Γ^D, Neumann boundary Γ^N and interior fault boundary Γ^F, with appropriate boundary conditions applied, the system can be summarized as follows:

$$\rho\frac{\partial^2 u_i}{\partial^2 t} - \frac{\partial \sigma_{ij}}{\partial x_j} - b_i = 0 \quad in \quad \Omega \tag{1}$$

$$\sigma_{ij} n_j = T_i \quad on \quad \Gamma^N \tag{2}$$

$$u_i = u_i^o \quad on \quad \Gamma^D \tag{3}$$

$$R_{ki}(u_i^+ - u_i^-) = \delta_k \quad on \quad \Gamma^F \tag{4}$$

Where b is the body force, δ is the imposed slip, R is the rotation matrix transforms the slip from global coordinate to local coordinate, $i, j = 1, 2, 3$ in three-dimension problem. We assume linear elastic material behavior, λ, μ are Lame constant, ϵ_{ij} is the infinitesimal strain:

$$\sigma_{ij} = \lambda\delta_{ij}\epsilon_{kk} + 2\mu\epsilon_{ij} \tag{5}$$

In antiplane (Mode III) problem, only the out-of-plane displacement $u = (0, 0, u(x_1, x_2))$ is the unknown variable, thus the equations above can be simplified:

$$\rho\frac{\partial^2 u_3}{\partial^2 t} = \frac{\partial}{\partial x_1}\Big(\frac{\partial \sigma_{13}}{\partial x_1}\Big) + \frac{\partial}{\partial x_2}\Big(\frac{\partial \sigma_{23}}{\partial x_2}\Big) = \frac{\partial}{\partial x_1}\Big(\mu\frac{\partial u_3}{\partial x_1}\Big) + \frac{\partial}{\partial x_2}\Big(\mu\frac{\partial u_3}{\partial x_2}\Big) + b_3 \quad in \quad \Omega \tag{6}$$

Here we assume quasi-dynamic by neglecting the inertia forces and body forces, so the governing equations reduce to Poisson equation, shown as follows:

$$\begin{aligned}
&\mu\Delta u_3 = \frac{\partial}{\partial x_1}\Big(\mu\frac{\partial u_3}{\partial x_1}\Big) + \frac{\partial}{\partial x_2}\Big(\mu\frac{\partial u_3}{\partial x_2}\Big) = 0 \\
&\sigma_{31} n_1 + \sigma_{32} n_2 = T_3 \quad on \quad \Gamma^N \\
&u_3 = u_3^o \quad on \quad \Gamma^D \\
&R_{k3}(u_3^+ - u_3^-) = \delta_k \quad on \quad \Gamma^F
\end{aligned} \tag{7}$$

1.2 Weak Form

Follow the unified approach in [1], we rewrite the equation (7) as a first-order system of variables $(u, \boldsymbol{\sigma})$ (we drop the index for simplicity):

$$
\begin{aligned}
\boldsymbol{\sigma} &= \mu \nabla u \\
-\nabla \cdot \boldsymbol{\sigma} &= 0 \qquad in \quad \Omega
\end{aligned}
\tag{8}
$$

Multiply the first and second equations by test functions $\boldsymbol{\tau}$ and v respectively, integrate on a subset K of Ω, we obtain the weak form:

$$
\begin{aligned}
\int_K \boldsymbol{\sigma} \cdot \boldsymbol{\tau} \, dx &= -\int_K \mu u \, \nabla \cdot \boldsymbol{\tau} \, dx + \int_{\partial K} \mu u \, n_K \cdot \boldsymbol{\tau} \, ds \\
\int_K \boldsymbol{\sigma} \cdot \nabla v \, dx &= \int_K f \, v \, dx + \int_{\partial K} \boldsymbol{\sigma} \cdot n_K \, v \, ds
\end{aligned}
\tag{9}
$$

1.3 Flux Form

In DG method, the boundary conditions at the element boundary ∂K is imposed weakly.

Here the finite element spaces defining $v \in V_h, \boldsymbol{\tau} \in \Sigma_h$ is given as follows:

$$
\begin{aligned}
V_h &:= \{v \in L^2(\Omega) \ : \ \hat{v}|_K \in P(K) \quad \forall K \in \mathcal{T}_h\}, \\
\tau_h &:= \{\boldsymbol{\tau} \in [L^2(\Omega)]^2 \ : \ \hat{\boldsymbol{\tau}}|_K \in \Sigma(K) \quad \forall K \in \mathcal{T}_h\}
\end{aligned}
\tag{10}
$$

Where $\mathcal{T}_h = \{K\}$ represents the triangulation of the domain Ω. The general DG formulation as an extension to equation (9), is to find $u_h \in V_h, \boldsymbol{\sigma_h} \in \mathcal{T}_h$, such that for all $K \in \mathcal{T}_h$ we have:

$$
\begin{aligned}
\int_K \boldsymbol{\sigma}_h \cdot \boldsymbol{\tau} \, dx &= -\int_K u_h \, \mu \nabla \cdot \boldsymbol{\tau} \, dx + \int_{\partial K} \mu \widehat{u}_K \, \mathbf{n}_K \cdot \boldsymbol{\tau} \, ds \quad \forall \boldsymbol{\tau} \in \Sigma(K), \\
\int_K \boldsymbol{\sigma}_h \cdot \nabla v \, dx &= \int_K f \, v \, dx + \int_{\partial K} \widehat{\boldsymbol{\sigma}}_K \cdot \mathbf{n}_K \, v \, ds \quad \forall v \in P(K),
\end{aligned}
\tag{11}
$$

The exposed numerical flux terms $(\hat{u}, \hat{\boldsymbol{\sigma}})$ need special form to be expressed in terms of $(u, \boldsymbol{\sigma})$ such that $\hat{u}$ approximates u and $\hat{\boldsymbol{\sigma}}$ approximates $\boldsymbol{\sigma} = \mu \nabla u$.

1.4 Primal Form

We define the set of facet: $\Gamma := \bigcup_{K \in \mathcal{T}_h} \partial K$ and different facet components: The faces with Dirichlet boundary condition Γ^D, interior faces are $\Gamma^i := \Gamma \backslash (\Gamma^D \cup \Gamma^N)$, the fault faces are Γ^F and the non-fault interior faces are $\Gamma^0 = \Gamma^i \backslash \Gamma^F$.

The other important definition is jump, average operators across two element shared boundary $(+, -)$. For ψ (scalar), $\boldsymbol{q}$ (vector), $\boldsymbol{w}$ (tensor), the definitions are summarized as follows:

$$
\begin{aligned}
\{\!\{\psi\}\!\} &= \frac{1}{2}\big(\psi^+ + \psi^-\big) \quad \text{(scalar)} \\
[\![\psi]\!] &= \psi^+ - \psi^- \quad \text{(scalar)}
\end{aligned}
\tag{12}
$$

$$
\begin{aligned}
\{\!\{\mathbf{q}\}\!\} &= \frac{1}{2}\big(\mathbf{q}^+ + \mathbf{q}^-\big) \quad \text{(vector)} \\
\{\!\{\mathbf{q} \cdot \mathbf{n}\}\!\} &= \frac{1}{2}\big(\mathbf{q}^+ \cdot \mathbf{n} + \mathbf{q}^- \cdot \mathbf{n}\big) \quad \text{(vector)} \\
[\![\mathbf{q}]\!] &= \mathbf{q}^+ - \mathbf{q}^- \quad \text{(vector)}
\end{aligned}
\tag{13}
$$

$$
\begin{aligned}
\{\!\{\mathbf{w}\}\!\} &= \frac{1}{2}\big(\mathbf{w}^+ + \mathbf{w}^-\big) \quad \text{(tensor)} \\
\{\!\{\mathbf{w}\cdot\mathbf{n}\}\!\} &= \frac{1}{2}\big(\mathbf{w}^+\cdot\mathbf{n} + \mathbf{w}^-\cdot\mathbf{n}\big) \quad \text{(vector)} \\
[\![\mathbf{w}]\!] &= \mathbf{w}^+ - \mathbf{w}^- \quad \text{(tensor)}
\end{aligned}
\tag{14}
$$

Following the steps in [1], we perform integration by part on equation (11):

$$
-\int_K u_h\,\mu\nabla\cdot\boldsymbol{\tau}\,dx = -\int_{\partial K} u_h\;\mu\;\boldsymbol{\tau}\;\boldsymbol{n}_K\,ds + \int_K \nabla u_h\,\mu\;\boldsymbol{\tau}\,dx \tag{15}
$$

Plug equation (15) into (11), we have the first equation:

$$
\int_K \boldsymbol{\sigma}_h\cdot\boldsymbol{\tau}\,dx = \int_K \nabla u_h\,\mu\;\boldsymbol{\tau}\,dx + \int_{\partial K}\mu\;(\widehat{u}_K - u_h)\,\mathbf{n}_K\cdot\boldsymbol{\tau}\,ds \quad \forall\boldsymbol{\tau}\in\Sigma(K) \tag{16}
$$

Note Identity in [1]:

> To deal with the sums of the form $\sum_{K\in\mathcal{T}_h}\int_{\partial K} q_K\varphi_K\cdot n_K\,ds$, we use the average and jump operators. A straightforward computation shows that for all $q\in T(\Gamma)$ and for all $\varphi\in[T(\Gamma)]^2$,
>
> $$\sum_{K\in\mathcal{T}_h}\int_{\partial K} q_K\varphi_K\cdot n_K\,ds = \int_\Gamma [\![q]\!]\cdot\{\varphi\}\,ds + \int_{\Gamma^0}\{q\}[\![\varphi]\!]\,ds. \tag{17}$$

Use the identity (1.4), we expand equation (18), and second equation in flux form (11) as follows:

$$
\int_K \boldsymbol{\sigma}_h\cdot\boldsymbol{\tau}\,dx = \int_K \nabla u_h\,\mu\;\boldsymbol{\tau}\,dx + \int_{\partial K}\mu\;[\![\widehat{u}_K - u_h]\!]\;\{\!\{\boldsymbol{\tau}\}\!\}\,ds + \int_{\partial K}\mu\;\{\!\{\widehat{u}_K - u_h\}\!\}\;[\![\boldsymbol{\tau}]\!]\,ds \tag{18}
$$

$$
\int_K \boldsymbol{\sigma}_h\cdot\nabla v\,dx = \int_K f\,v\,dx + \int_{\partial K}[\![\widehat{\boldsymbol{\sigma}}_K]\!]\;\{\!\{v\}\!\}\,ds + \int_{\partial K}\{\!\{\widehat{\boldsymbol{\sigma}}_K\}\!\}\;[\![v]\!]\,ds \tag{19}
$$

Define lifting operators as in [1]:

> Recalling that $\nabla_h V_h\subset\Sigma_h$ and defining lifting operators $r:[L^2(\Gamma)]^2\to\Sigma_h$ and $l: L^2(\Gamma^0)\to\Sigma_h$ by
>
> $$\int_\Omega r(\varphi)\cdot\tau\,dx = -\int_\Gamma \varphi\cdot\{\!\{\tau\}\!\}\,ds, \qquad \int_\Omega l(q)\cdot\tau\,dx = -\int_{\Gamma^0} q[\![\tau]\!]\,ds \quad \forall\tau\in\Sigma_h, \tag{20}$$

Then we could rewrite $\boldsymbol{\sigma}_h$ as follows:

$$
\boldsymbol{\sigma}_h = \mu\Big(\nabla u_h - r([\![\widehat{u}_K - u_h]\!]) - l(\{\!\{\widehat{u}_K - u_h\}\!\})\Big) \tag{21}
$$

Now assert $\boldsymbol{\tau}=\nabla v$, plug equation (21) into second equation in (19):

$$
B_h(u_h, v) = \int_\Omega f v\,dx \quad \forall v\in V_h, \tag{22}
$$

where

$$
\begin{aligned}
B_h(u_h, v) := \int_\Omega \mu \nabla_h u_h \cdot \nabla_h v \, dx + \int_\Gamma \big(\mu [\![\hat{u} - u_h]\!] \cdot \{\nabla_h v\} - \{\hat{\boldsymbol{\sigma}}\} \cdot [\![v]\!]\big) \, ds \\
+ \int_{\Gamma^i} \big(\mu \{\hat{u} - u_h\} [\![\nabla_h v]\!] - [\![\hat{\boldsymbol{\sigma}}]\!] \{v\}\big) \, ds.
\end{aligned}
\tag{23}
$$

1.5 Types of numerical flux

In this study, we implement two types of numerical flux: BR2 (Bassi-Rebay 2) and IP (Interior Penalty).

1.5.1 Bassi-Rebay 2 (BR2)

The numerical flux choice:

$$
\hat{u} = \{\!\{u_h\}\!\}; \quad \hat{\boldsymbol{\sigma}} = \{\!\{\boldsymbol{\sigma_h}\}\!\} - \alpha_r([\![u_h]\!]) \tag{24}
$$

The final bilinear form $B_h(u_h, v)$ is given as follows, see Appendix for derivation (need to be done!):

$$
B_h(u_h, v) = \int_\Omega \mu \nabla_h u_h \cdot \nabla_h v \, dx - \int_\Gamma \mu \big([\![u_h]\!]_a \cdot \{\nabla_h v\} + \{\nabla_h u_h\} \cdot [\![v]\!]_a\big) \, ds + \mu \alpha^r(u_h, v) \tag{25}
$$

where

$$
\alpha^r(u_h, v) = \int_\Gamma \alpha_r(u_h) \cdot [\![v]\!]_a \, ds = \sum_{e \in \mathcal{E}_h} \int_\Omega \eta_e \, r_e([\![u_h]\!]_a) \cdot r_e([\![v]\!]_a) \, dx. \tag{26}
$$

Note there is a definition convention between Arnold's jump definition $[\![u]\!]_a$ and our current definition $[\![u]\!]$:

$$
\{\!\{\nabla u\}\!\} \cdot [\![v]\!]_a = \{\!\{\nabla u\}\!\} \cdot [\![v \cdot n]\!] = \{\!\{\nabla u \cdot n\}\!\} \cdot [\![v]\!] \tag{27}
$$

The bilinear form that agrees with implementation is as follows:

$$
B_h(u_h, v) = \int_\Omega \mu \nabla_h u_h \cdot \nabla_h v \, dx - \int_\Gamma \mu \big([\![u_h]\!] \cdot \{\!\{\nabla_h v \cdot n\}\!\} + \{\!\{\nabla_h u_h \cdot n\}\!\} \cdot [\![v]\!]\big) \, ds + \mu \alpha^r(u_h, v) \tag{28}
$$

where

$$
\alpha^r(u_h, v) = \int_\Gamma \alpha_r(u_h) \cdot [\![v]\!] \, ds = \sum_{e \in \mathcal{E}_h} \int_\Omega \eta_e \, r_e([\![u_h]\!]) \cdot r_e([\![v]\!]) \, dx. \tag{29}
$$

Note the BR2 lifting is a volume integral over two adjacent elements.

Expanding the $\Gamma = \Gamma^D + \Gamma^F + \Gamma^0 + \Gamma^N$, we have the full expansion of bilinear form $B_h(u_h, v)$, where the displacement jump δ is enforced on Γ^F weakly, terms with Γ^N is neglected by traction-free Neumann boundary conditions:

$$
\begin{aligned}
B_h(u_h, v_h) = &\int_\Omega \mu \nabla u_h \cdot \nabla v_h \, dx \\
&- \sum_{e \in \Gamma^0 \cup \Gamma^F} \int_e \{\!\{ \mu \nabla u_h \cdot \mathbf{n} \}\!\} [\![v_h]\!] \, ds \\
&- \sum_{e \in \Gamma^0 \cup \Gamma^F} \int_e \{\!\{ \mu \nabla v_h \cdot \mathbf{n} \}\!\} [\![u_h]\!] \, ds \\
&+ \sum_{e \in \Gamma^0 \cup \Gamma^F} \int_\Omega \eta_e \mu \, r_e([\![u_h]\!]) \cdot r_e([\![v_h]\!]) \, dx \\
&- \sum_{e \in \Gamma^D} \int_e (\mu \nabla u_h \cdot \mathbf{n}) \, v_h \, ds \\
&- \sum_{e \in \Gamma^D} \int_e (\mu \nabla v_h \cdot \mathbf{n}) \, u_h \, ds \\
&+ \sum_{e \in \Gamma^D} \int_\Omega \eta_e \mu \, r_e(u_h) \cdot r_e(v_h) \, dx
\end{aligned} \tag{30}
$$

$$
\begin{aligned}
L(v_h) = &- \sum_{e \in \Gamma^D} \int_e (\mu \nabla v_h \cdot \mathbf{n}) \, u^o \, ds \\
&+ \sum_{e \in \Gamma^D} \int_\Omega \eta_e \mu \, r_e(u^o) \cdot r_e(v_h) \, dx \\
&- \sum_{e \in \Gamma^F} \int_e \{\!\{ \mu \nabla v_h \cdot \mathbf{n} \}\!\} \, \delta \, ds \\
&+ \sum_{e \in \Gamma^F} \int_\Omega \eta_e \mu \, r_e(\delta) \cdot r_e([\![v_h]\!]) \, dx
\end{aligned} \tag{31}
$$

1.5.2 Interior Penalty (IP)

The numerical flux choice:

$$\hat{u} = \{\!\{ u_h \}\!\}; \quad \hat{\boldsymbol{\sigma}} = \mu\Big(\{\nabla_h u_h\} - \alpha_j([\![u_h]\!]) \Big) \tag{32}$$

The final bilinear form $B_h(u_h, v)$ is given as follows, see Appendix for derivation (need to be done!):

$$B_h(u_h, v) = \int_\Omega \mu \nabla_h u_h \cdot \nabla_h v \, dx - \int_\Gamma \mu\big([\![u_h]\!]_a \cdot \{\nabla_h v\} + \{\nabla_h u_h\} \cdot [\![v]\!]_a \big) \, ds + \mu \alpha^j(u_h, v) \tag{33}$$

where

$$\alpha^j(u_h, v) = \int_\Gamma \frac{\eta_e}{h_e} [\![u_h]\!] \cdot [\![v]\!] \, ds \tag{34}$$

Note there is a definition convention between Arnold's jump definition $[\![u]\!]_a$ and our current definition $[\![u]\!]$:

$$\{\!\{ \nabla u \}\!\} \cdot [\![v]\!]_a = \{\!\{ \nabla u \}\!\} \cdot [\![v \cdot n]\!] = \{\!\{ \nabla u \cdot n \}\!\} \cdot [\![v]\!] \tag{35}$$

The bilinear form that agrees with implementation is as follows:

$$B_h(u_h, v) = \int_\Omega \mu \nabla_h u_h \cdot \nabla_h v \, dx - \int_\Gamma \mu \big([\![u_h]\!] \cdot \{\nabla_h v \cdot n\} + \{\nabla_h u_h \cdot n\} \cdot [\![v]\!] \big) \, ds + \mu \alpha^j(u_h, v) \tag{36}$$

where

$$\alpha^j(u_h, v) = \int_\Gamma \frac{\eta_e}{h_e} [\![u_h]\!] \cdot [\![v]\!] \, ds \tag{37}$$

Note unlike BR2, the penalty term is a surface integral.

Expanding the $\Gamma = \Gamma^D + \Gamma^F + \Gamma^0 + \Gamma^N$, we have the full expansion of bilinear form $B_h(u_h, v)$, where the displacement jump δ is enforced on Γ^F weakly, terms with Γ^N is neglected by traction-free Neumann boundary conditions:

$$\boxed{\begin{aligned}
B_h(u_h, v_h) = & \int_\Omega \mu \nabla u_h \cdot \nabla v_h \, dx \\
& - \sum_{e \in \Gamma^0 \cup \Gamma^F} \int_e \{\!\{\mu \nabla u_h \cdot \mathbf{n}\}\!\} [\![v_h]\!] \, ds \\
& - \sum_{e \in \Gamma^0 \cup \Gamma^F} \int_e \{\!\{\mu \nabla v_h \cdot \mathbf{n}\}\!\} [\![u_h]\!] \, ds \\
& + \sum_{e \in \Gamma^0 \cup \Gamma^F} \int_e \mu \frac{\eta_e}{h_e} [\![u_h]\!] \cdot [\![v_h]\!] \, ds \qquad (38) \\
& - \sum_{e \in \Gamma^D} \int_e (\mu \nabla u_h \cdot \mathbf{n}) \, v_h \, ds \\
& - \sum_{e \in \Gamma^D} \int_e (\mu \nabla v_h \cdot \mathbf{n}) \, u_h \, ds \\
& + \sum_{e \in \Gamma^D} \int_e \mu \frac{\eta_e}{h_e} u_h \; v_h \; ds \\[2ex]
L(v_h) = & - \sum_{e \in \Gamma^D} \int_e (\mu \nabla v_h \cdot \mathbf{n}) \, u^o \, ds \\
& + \sum_{e \in \Gamma^D} \int_e \mu \frac{\eta_e}{h_e} u^o \cdot [\![v_h]\!] \, ds \\
& - \sum_{e \in \Gamma^F} \int_e \{\!\{\mu \nabla v_h \cdot \mathbf{n}\}\!\} \, \delta \, ds \qquad (39) \\
& + \sum_{e \in \Gamma^F} \int_e \mu \frac{\eta_e}{h_e} \delta \cdot [\![v_h]\!] \, ds
\end{aligned}}$$

1.6 Rate-and-State Friction

The rate-and-state friction follows BP2 Benchmark description, shear stress τ on the fault is equal to fault strength F:

$$\tau = F(V, \theta) + \eta V \tag{40}$$

Where $\tau = \tau^0 + \tau^{qs}$ is the sum of prestress, quasi-static shear stress solution and ηV is the radiation damping approximation to inertia, $\eta = \mu/(2c_s)$, where c_s is the shear wave speed.

The fault strength $F(V, \theta)$ is a function of slip rate V and state variable θ:

$$F(V, \theta) = \sigma_n f(V, \theta) \tag{41}$$

Where σ_n is normal stress and θ evolves according to the aging law:

$$\frac{d\theta}{dt} = 1 - \frac{V\theta}{D_c} \tag{42}$$

Where D_c is the critical slip distance. The friction coefficient f is given by a regularized formulation:

$$f(V, \theta) = a \sinh^{-1}\left[\frac{V}{2V_0} \exp\left(\frac{f_0 + b\ln(V_0\theta/D_\text{c})}{a}\right)\right] \tag{43}$$

1.7 Algorithm

Input: Slip $\{\delta_i^n\}$, state variable $\{\theta_i^n\}$, time t^n, step size Δt^n
Output: Updated $\{\delta_i^{n+1}\}$, $\{\theta_i^{n+1}\}$, time t^{n+1}, adapted Δt^{n+1}

% — Initialization (first step only) —

if $n = 0$ **then**

  Set $\delta_i = 0$, $\quad \theta_i = D_c/V_\text{init}$ for all fault nodes i

  Solve $B_h(u_h, v_h) = L_h(v_h;\, \boldsymbol{\delta} = \mathbf{0})$ for u_h ▷ DG elastic solve

  Extract $\tau_{\text{qs},i} = \{\!\{\mu\, \nabla u_h \cdot \hat{\mathbf{n}}\}\!\}\big|_i$ ▷ Average flux on fault face

  Compute uniform pre-stress (constant, evaluated at steady state with $a = a_\text{max}$):

$$\tau_0 = \sigma_n\, a_\text{max} \sinh^{-1}\left[\frac{V_\text{init}}{2V_0} \exp\left(\frac{f_0 + b\ln(V_0/V_\text{init})}{a_\text{max}}\right)\right] + \eta\, V_\text{init}$$

  Compute equilibrium friction: $f_i^* = (\tau_0 + \tau_{\text{qs},i} - \eta_i V_\text{init})/\sigma_n$

  Initialize state variable:

$$\theta_i^0 = \frac{D_c}{V_0} \exp\left[\frac{a_i}{b} \ln\left(\frac{2V_0}{V_\text{init}} \sinh\left(\frac{f_i^*}{a_i}\right)\right) - \frac{f_0}{b}\right]$$

end if

% — Dormand–Prince RK45: 7 stages per step (FSAL) —

for each RK stage $s = 1, \ldots, 7$ **do**

  % Step 1: Assemble stage values from RK tableaux

  $\tilde{\delta}_i = \delta_i^n + \Delta t^n \sum_j a_{sj}\, k_j^{\delta,i}, \quad \tilde{\theta}_i = \theta_i^n + \Delta t^n \sum_j a_{sj}\, k_j^{\theta,i}$

  % Step 2: Elastic solve

  Solve $B_h(u_h, v_h) = L_h(v_h;\, \tilde{\boldsymbol{\delta}})$ for u_h ▷ BR2/IP; slip via Nitsche

  % Step 3: Extract traction and compute total stress

  $\tau_{\text{qs},i} = \{\!\{\mu\, \nabla u_h \cdot \hat{\mathbf{n}}\}\!\}\big|_i$ ▷ Average flux on fault face

  $\tau_i = \tau_0 + \tau_{\text{qs},i}$

  % Step 4: Solve for slip rate (seismogenic zone)

  for each fault node i with $z_i \geq -W_f$ **do**

Solve for V_i via Newton–Raphson:

$$\sigma_n \, f(V_i, \, \tilde{\theta}_i) = \tau_i - \eta_i \, V_i, \qquad \eta_i = \frac{\mu}{2c_s}$$

repeat

$\xi \leftarrow \dfrac{V_i}{2V_0} \exp\left(\dfrac{f_0 + b \ln(V_0 \tilde{\theta}_i / D_c)}{a_i}\right)$

$f \leftarrow a_i \sinh^{-1}(\xi)$

$F \leftarrow \sigma_n f + \eta_i V_i - \tau_i$

$F' \leftarrow \sigma_n \dfrac{a_i}{2V_0 \sqrt{1 + \xi^2}} \exp\left(\dfrac{f_0 + b \ln(V_0 \tilde{\theta}_i / D_c)}{a_i}\right) + \eta_i$

$V_i \leftarrow V_i - F/F'$ ▷ With bisection safeguard for bounds

until $|V_i^{\text{new}} - V_i| / \max(|V_i|, V_{\min}) < 10^{-12}$ **or** $|F| < 10^{-12} \, \tau_i$

end for

% Step 5: Compute ODE rates

for each fault node i **do**

if $z_i \geq -W_f$ **then** ▷ Seismogenic zone

$k_s^{\delta,i} = V_i$

$k_s^{\theta,i} = 1 - V_i \, \tilde{\theta}_i / D_c$ ▷ Aging law

else ▷ Below seismogenic zone

$k_s^{\delta,i} = V_p$ ▷ Prescribed plate rate

$k_s^{\theta,i} = 0$

end if

end for

end for

% — Step 6: Update solution and adapt time step —

$\delta_i^{n+1} = \delta_i^n + \Delta t^n \sum_{s=1}^{6} b_s \, k_s^{\delta,i}, \quad \theta_i^{n+1} = \theta_i^n + \Delta t^n \sum_{s=1}^{6} b_s \, k_s^{\theta,i}$ ▷ b_s: 5th-order weights; $b_2 = b_7 = 0$

Compute error estimate per state component: $e_i = \Delta t^n \sum_{s=1}^{7} (b_s - \hat{b}_s) \, k_s^i, \quad i \in \{\delta_0, \theta_0, \delta_1, \theta_1, \ldots\}$

▷ $\hat{b}_s$: 4th-order weights

Compute scaled error norm: $\epsilon = \max_i \dfrac{|e_i|}{\texttt{atol} + \texttt{rtol} \cdot |y_i^{n+1}|}$ ▷ Global reduction across MPI ranks

if $\epsilon \leq 1$ **then** accept step; **else** reject and retry with reduced Δt

Adapt $\Delta t^{n+1} = \min\Big(\Delta t_{\max}, \; \max\big(\Delta t_{\min}, \; \gamma \, \Delta t^n \, \epsilon^{-1/5}\big)\Big)$ ▷ $\gamma = 0.9$ safety factor

$t^{n+1} = t^n + \Delta t^n$

1.8 Verification: SCEC-SEAS-BP1

2 Quasi-Dynamic 3D Half Space Problem

2.1 Strong Form

The governing equations are defined in the physical domain Ω with its boundary $\partial\Omega$. The boundary $\partial\Omega$ is partitioned into a Dirichlet boundary Γ^D, a Neumann boundary Γ^N, and an interior fault boundary Γ^F. We assume linear elastic material behavior with the constitutive relation $\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}$, where $\mathbb{C}$ is the fourth-order stiffness tensor and $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T)$ is the infinitesimal strain tensor. For isotropic materials, $\mathbb{C}$ is characterized by the Lamé constants λ and μ, i.e., $C_{ijkl} = \lambda \delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$. With appropriate boundary conditions applied, the system can be summarized as follows:

$$
\begin{aligned}
\rho \ddot{\boldsymbol{u}} - \nabla \cdot \boldsymbol{\sigma} &= \boldsymbol{b} && \text{in} \quad \Omega \\
\boldsymbol{\sigma} \boldsymbol{n} &= \boldsymbol{T} && \text{on} \quad \Gamma^N \\
\boldsymbol{u} &= \boldsymbol{g}^D && \text{on} \quad \Gamma^D \\
\boldsymbol{R} [\![\boldsymbol{u}]\!] &= \boldsymbol{\delta} && \text{on} \quad \Gamma^F
\end{aligned}
\tag{44}
$$

where $\boldsymbol{b}$ is the body-force vector, $\boldsymbol{T}$ is the prescribed traction, $\boldsymbol{g}^D$ is the prescribed displacement, $\boldsymbol{\delta}$ is the imposed slip, $\boldsymbol{R}$ is the rotation matrix that transforms the displacement jump from global to local (fault-aligned) coordinates, and $[\![\boldsymbol{u}]\!] = \boldsymbol{u}^- - \boldsymbol{u}^+$ is the displacement jump across the fault. In indicial notation, the governing equation and constitutive relation read:

$$
\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial \sigma_{ij}}{\partial x_j} = b_i, \qquad \sigma_{ij} = C_{ijkl}\, \varepsilon_{kl} \tag{45}
$$

where $i, j, k, l = 1, \dots, D$ with D being the spatial dimension.

2.2 Weak Form

Following the unified approach in [1], we rewrite equation (44) as a first-order system of variables $(\boldsymbol{u}, \boldsymbol{\sigma})$. For the quasi-static case ($\rho \ddot{\boldsymbol{u}} = \boldsymbol{0}$), we drop the inertia term and write (omitting the index for simplicity):

$$
\begin{aligned}
\boldsymbol{\sigma} &= \mathbb{C} : \boldsymbol{\varepsilon} \\
-\nabla \cdot \boldsymbol{\sigma} &= \boldsymbol{b} \qquad \text{in} \quad \Omega
\end{aligned}
\tag{46}
$$

where $\boldsymbol{b}$ is the body-force vector.

Multiply the first and second equations by test functions $\boldsymbol{\tau}$ and $\boldsymbol{v}$ respectively, integrate on a subset K of Ω, we obtain the weak form:

$$
\int_K \boldsymbol{\tau} : \boldsymbol{\sigma}\, dx - \int_K \boldsymbol{\tau} : \mathbb{C} : \nabla \boldsymbol{u}\, dx = 0 \tag{47}
$$

$$
\int_K \nabla \boldsymbol{v} : \boldsymbol{\sigma}\, dx - \int_{\partial K} \boldsymbol{v} \cdot \boldsymbol{\sigma} \boldsymbol{n}\, ds = \int_K \boldsymbol{v} \cdot \boldsymbol{b}\, dx \tag{48}
$$

where the second equation is obtained by integration by parts.

2.3 Flux Form

From equation (47), to transfer the gradient from $\boldsymbol{u}$ onto $\boldsymbol{\tau} : \mathbb{C}$, we use the product rule identity:

$$
\boldsymbol{\tau} : \mathbb{C} : \nabla \boldsymbol{u} = \nabla \cdot \left[(\boldsymbol{\tau} : \mathbb{C})^T \boldsymbol{u} \right] - (\nabla \cdot (\boldsymbol{\tau} : \mathbb{C})) \cdot \boldsymbol{u} \tag{49}
$$

Here $\boldsymbol{\tau} : \mathbb{C}$ is a second-order tensor obtained by contracting $\boldsymbol{\tau}$ with $\mathbb{C}$, and $(\boldsymbol{\tau} : \mathbb{C})^T \boldsymbol{u}$ is a vector, to which the divergence theorem applies. Substituting (49) into (47):

$$\int_K \boldsymbol{\tau} : \boldsymbol{\sigma}\, dx = \int_K \nabla \cdot \left[(\boldsymbol{\tau} : \mathbb{C})^T \boldsymbol{u}\right] dx - \int_K (\nabla \cdot (\boldsymbol{\tau} : \mathbb{C})) \cdot \boldsymbol{u}\, dx \tag{50}$$

Applying the divergence theorem to the first term on the right-hand side:

$$\int_K \nabla \cdot \left[(\boldsymbol{\tau} : \mathbb{C})^T \boldsymbol{u}\right] dx = \int_{\partial K} \boldsymbol{u} \cdot (\boldsymbol{\tau} : \mathbb{C})\boldsymbol{n}\, ds \tag{51}$$

where we have used the symmetry of $\mathbb{C}$ to drop the transpose. Rearranging:

$$\int_K \boldsymbol{\tau} : \boldsymbol{\sigma}\, dx + \int_K (\nabla \cdot (\boldsymbol{\tau} : \mathbb{C})) \cdot \boldsymbol{u}\, dx - \int_{\partial K} \boldsymbol{u} \cdot (\boldsymbol{\tau} : \mathbb{C})\boldsymbol{n}\, ds = 0 \tag{52}$$

In the DG framework, $\boldsymbol{u}$ is double-valued on element boundaries. We replace the trace of $\boldsymbol{u}$ on ∂K with a numerical flux $\hat{\boldsymbol{u}}$:

$$\boxed{\int_K \boldsymbol{\tau} : \boldsymbol{\sigma}\, dx + \int_K (\nabla \cdot (\boldsymbol{\tau} : \mathbb{C})) \cdot \boldsymbol{u}\, dx - \int_{\partial K} \hat{\boldsymbol{u}} \cdot (\boldsymbol{\tau} : \mathbb{C})\boldsymbol{n}\, ds = 0} \tag{53}$$

Then for equation (48), similarly, $\boldsymbol{\sigma}$ is double-valued on element boundaries. We replace the traction $\boldsymbol{\sigma n}$ on ∂K with a numerical flux $\hat{\boldsymbol{\sigma}}\boldsymbol{n}$:

$$\boxed{\int_K \nabla \boldsymbol{v} : \boldsymbol{\sigma}\, dx - \int_{\partial K} \boldsymbol{v} \cdot \hat{\boldsymbol{\sigma}}\boldsymbol{n}\, ds = \int_K \boldsymbol{v} \cdot \boldsymbol{b}\, dx} \tag{54}$$

Equations (53) and (54) constitute the *flux form* of the DG discretization. The specific DG method (e.g., SIPG, NIPG, or others) is determined by the choice of the numerical fluxes $\hat{\boldsymbol{u}}$ and $\hat{\boldsymbol{\sigma}}$ on interior faces, fault faces, and external boundaries.

2.4 Primal Form

2.4.1 Bassi-Rebay 2 (BR2)

Starting from the flux form derived previously, we now eliminate $\boldsymbol{\sigma}$ and arrive at the primal formulation.

Recovering $\boldsymbol{\sigma}$ in terms of $\boldsymbol{u}$ and the displacement flux. We reverse the integration by parts in (53). Applying the product rule identity (49) in the opposite direction:

$$\int_K (\nabla \cdot (\boldsymbol{\tau} : \mathbb{C})) \cdot \boldsymbol{u}\, dx = -\int_K \boldsymbol{\tau} : \mathbb{C} : \nabla \boldsymbol{u}\, dx + \int_{\partial K} \boldsymbol{u} \cdot (\boldsymbol{\tau} : \mathbb{C})\boldsymbol{n}\, ds \tag{55}$$

Substituting back into (53):

$$\int_K \boldsymbol{\tau} : \boldsymbol{\sigma}\, dx = \int_K \boldsymbol{\tau} : \mathbb{C} : \nabla \boldsymbol{u}\, dx + \int_{\partial K} (\hat{\boldsymbol{u}} - \boldsymbol{u}) \cdot (\boldsymbol{\tau} : \mathbb{C})\boldsymbol{n}\, ds \tag{56}$$

Setting $\boldsymbol{\tau} = \nabla \boldsymbol{v}$. Since $\nabla_h V_h \subset \Sigma_h$, we may choose $\boldsymbol{\tau} = \nabla \boldsymbol{v}$ in (56):

$$\int_K \nabla \boldsymbol{v} : \boldsymbol{\sigma}\, dx = \int_K \nabla \boldsymbol{v} : \mathbb{C} : \nabla \boldsymbol{u}\, dx + \int_{\partial K} (\hat{\boldsymbol{u}} - \boldsymbol{u}) \cdot (\nabla \boldsymbol{v} : \mathbb{C})\boldsymbol{n}\, ds \tag{57}$$

Substituting (57) into (54) eliminates $\boldsymbol{\sigma}$:

$$\int_K \nabla \boldsymbol{v} : \mathbb{C} : \nabla \boldsymbol{u}\, dx + \int_{\partial K} (\hat{\boldsymbol{u}} - \boldsymbol{u}) \cdot (\nabla \boldsymbol{v} : \mathbb{C})\boldsymbol{n}\, ds - \int_{\partial K} \boldsymbol{v} \cdot \hat{\boldsymbol{\sigma}}\boldsymbol{n}\, ds = \int_K \boldsymbol{v} \cdot \boldsymbol{b}\, dx \tag{58}$$

Averages and jumps. Let K^+ and K^- be two elements sharing a face $e = \partial K^+ \cap \partial K^-$, with $\boldsymbol{n}$ pointing from K^- to K^+. For a vector $\boldsymbol{q}$ and tensor $\boldsymbol{w}$:

$$\{\!\{\boldsymbol{q}\}\!\} = \frac{1}{2}(\boldsymbol{q}^+ + \boldsymbol{q}^-), \qquad [\![\boldsymbol{q}]\!] = \boldsymbol{q}^- - \boldsymbol{q}^+ \tag{59}$$

and similarly for tensors. Note the convention $[\![\cdot]\!] = (\cdot)^- - (\cdot)^+$, consistent with [**uphoff2023**].

Recap of the lifting operators. Recall the lifting operators $\boldsymbol{r} : [L^2(\Gamma)]^2 \to \Sigma_h$ and $l : L^2(\Gamma^0) \to \Sigma_h$ defined by

$$\int_\Omega \boldsymbol{r}(\boldsymbol{\varphi}) \cdot \boldsymbol{\tau}\, dx = -\int_\Gamma \boldsymbol{\varphi} \cdot \{\!\{\boldsymbol{\tau}\}\!\}\, ds, \qquad \int_\Omega l(q) \cdot \boldsymbol{\tau}\, dx = -\int_{\Gamma^0} q[\![\boldsymbol{\tau}]\!]\, ds \qquad \forall \boldsymbol{\tau} \in \Sigma_h \tag{20}$$

For the BR2 method, we define the *local* lifting operator $\boldsymbol{r}_e : [L^2(e)]^2 \to \Sigma_h$ restricted to a single face e:

$$\int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{\varphi}) \cdot \boldsymbol{\tau}\, dx = -\int_e \boldsymbol{\varphi} \cdot \{\!\{\boldsymbol{\tau}\}\!\}\, ds \qquad \forall \boldsymbol{\tau} \in \Sigma_h \tag{60}$$

where $\omega_e = K^- \cup K^+$ is the patch of the two elements sharing face e (or the single element adjacent to a boundary face).

Definition of the BR2 numerical fluxes. On an *interior face* $e \in \Gamma_0$:

$$\begin{aligned} \hat{\boldsymbol{\sigma}}\boldsymbol{n} &= \{\!\{\mathbb{C} : \nabla \boldsymbol{u}\}\!\}\boldsymbol{n} + \eta_e \{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!])\}\!\}\boldsymbol{n} \\ \hat{\boldsymbol{u}} &= \{\!\{\boldsymbol{u}\}\!\} \end{aligned} \tag{61}$$

where $\eta_e \geq 1$ is a stabilization parameter, typically set to $\eta_e = D + 1$ for simplices. On a *fault face* $e \in \Gamma_F$, the fault condition $[\![\boldsymbol{u}]\!] = \boldsymbol{g}^F$ is enforced through a double-valued displacement flux:

$$\begin{aligned} \hat{\boldsymbol{\sigma}}\boldsymbol{n} &= \{\!\{\mathbb{C} : \nabla \boldsymbol{u}\}\!\}\boldsymbol{n} + \eta_e \{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!] - \boldsymbol{g}^F)\}\!\}\boldsymbol{n} \\ \hat{\boldsymbol{u}} &= \{\!\{\boldsymbol{u}\}\!\} \quad \text{(double-valued: } \hat{\boldsymbol{u}}|_{K^-} = \{\!\{\boldsymbol{u}\}\!\} + \boldsymbol{g}^F/2,\ \hat{\boldsymbol{u}}|_{K^+} = \{\!\{\boldsymbol{u}\}\!\} - \boldsymbol{g}^F/2) \end{aligned} \tag{62}$$

so that $[\![\hat{\boldsymbol{u}}]\!] = \hat{\boldsymbol{u}}|_{K^-} - \hat{\boldsymbol{u}}|_{K^+} = \boldsymbol{g}^F$. On a *Dirichlet boundary* $e \in \Gamma_D$:

$$\begin{aligned} \hat{\boldsymbol{\sigma}}\boldsymbol{n} &= (\mathbb{C} : \nabla \boldsymbol{u})\boldsymbol{n} + \eta_e\, \boldsymbol{r}_e(\boldsymbol{u} - \boldsymbol{g}^D)\boldsymbol{n} \\ \hat{\boldsymbol{u}} &= \boldsymbol{g}^D \end{aligned} \tag{63}$$

On a *Neumann boundary* $e \in \Gamma_N$:

$$\begin{aligned} \hat{\boldsymbol{\sigma}}\boldsymbol{n} &= \boldsymbol{0} \\ \hat{\boldsymbol{u}} &= \boldsymbol{u} \end{aligned} \tag{64}$$

Inserting the fluxes into the primal form. We substitute the BR2 fluxes into (58) and sum over all elements. We label the four terms:

$$\underbrace{\int_K \nabla \boldsymbol{v} : \mathbb{C} : \nabla \boldsymbol{u}\, dx}_{\text{Term I}} + \underbrace{\int_{\partial K} (\hat{\boldsymbol{u}} - \boldsymbol{u}) \cdot (\nabla \boldsymbol{v} : \mathbb{C})\boldsymbol{n}\, ds}_{\text{Term II}} - \underbrace{\int_{\partial K} \boldsymbol{v} \cdot \hat{\boldsymbol{\sigma}}\boldsymbol{n}\, ds}_{\text{Term III}} = \underbrace{\int_K \boldsymbol{v} \cdot \boldsymbol{b}\, dx}_{\text{Term IV}} \tag{58}$$

Recall: outward normal of K^- on e is $+\boldsymbol{n}$; outward normal of K^+ on e is $-\boldsymbol{n}$.

Term I: Volume integral.

$$\text{(I)} = \sum_{K \in \mathcal{T}_h} \int_K \nabla \boldsymbol{v} : \mathbb{C} : \nabla \boldsymbol{u}\, dx \tag{65}$$

Term II: Boundary correction (displacement flux). *Interior faces ($e \in \Gamma_0$):* Here $\hat{\boldsymbol{u}} = \{\!\{\boldsymbol{u}\}\!\}$. The two elements contribute:

$$\int_e (\{\!\{\boldsymbol{u}\}\!\} - \boldsymbol{u}^-) \cdot (\nabla \boldsymbol{v}^- : \mathbb{C})(+\boldsymbol{n})\, ds + \int_e (\{\!\{\boldsymbol{u}\}\!\} - \boldsymbol{u}^+) \cdot (\nabla \boldsymbol{v}^+ : \mathbb{C})(-\boldsymbol{n})\, ds \tag{66}$$

Using $[\![\boldsymbol{u}]\!] = \boldsymbol{u}^- - \boldsymbol{u}^+$:

$$\{\!\{\boldsymbol{u}\}\!\} - \boldsymbol{u}^- = -\tfrac{1}{2}[\![\boldsymbol{u}]\!], \qquad \{\!\{\boldsymbol{u}\}\!\} - \boldsymbol{u}^+ = +\tfrac{1}{2}[\![\boldsymbol{u}]\!] \tag{67}$$

Substituting:

$$\begin{aligned} &= \int_e (-\tfrac{1}{2}[\![\boldsymbol{u}]\!]) \cdot (\nabla \boldsymbol{v}^- : \mathbb{C})\boldsymbol{n}\, ds + \int_e (+\tfrac{1}{2}[\![\boldsymbol{u}]\!]) \cdot (\nabla \boldsymbol{v}^+ : \mathbb{C})(-\boldsymbol{n})\, ds \\ &= -\int_e \tfrac{1}{2}[\![\boldsymbol{u}]\!] \cdot (\nabla \boldsymbol{v}^- : \mathbb{C})\boldsymbol{n}\, ds - \int_e \tfrac{1}{2}[\![\boldsymbol{u}]\!] \cdot (\nabla \boldsymbol{v}^+ : \mathbb{C})\boldsymbol{n}\, ds \\ &= -\int_e [\![\boldsymbol{u}]\!] \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{v}\}\!\}\boldsymbol{n}\, ds \end{aligned} \tag{68}$$

Fault faces ($e \in \Gamma_F$): The displacement flux is double-valued: $\hat{\boldsymbol{u}}|_{K^-} = \{\!\{\boldsymbol{u}\}\!\} + \boldsymbol{g}^F/2$ and $\hat{\boldsymbol{u}}|_{K^+} = \{\!\{\boldsymbol{u}\}\!\} - \boldsymbol{g}^F/2$. Using $\{\!\{\boldsymbol{u}\}\!\} - \boldsymbol{u}^- = -\frac{1}{2}[\![\boldsymbol{u}]\!]$ and $\{\!\{\boldsymbol{u}\}\!\} - \boldsymbol{u}^+ = +\frac{1}{2}[\![\boldsymbol{u}]\!]$:

$$\begin{aligned} &= \int_e (-\tfrac{1}{2}[\![\boldsymbol{u}]\!] + \tfrac{1}{2}\boldsymbol{g}^F) \cdot (\nabla \boldsymbol{v}^- : \mathbb{C})\boldsymbol{n}\, ds + \int_e (+\tfrac{1}{2}[\![\boldsymbol{u}]\!] - \tfrac{1}{2}\boldsymbol{g}^F) \cdot (\nabla \boldsymbol{v}^+ : \mathbb{C})(-\boldsymbol{n})\, ds \\ &= \int_e (-\tfrac{1}{2}[\![\boldsymbol{u}]\!] + \tfrac{1}{2}\boldsymbol{g}^F) \cdot (\nabla \boldsymbol{v}^- : \mathbb{C})\boldsymbol{n}\, ds + \int_e (-\tfrac{1}{2}[\![\boldsymbol{u}]\!] + \tfrac{1}{2}\boldsymbol{g}^F) \cdot (\nabla \boldsymbol{v}^+ : \mathbb{C})\boldsymbol{n}\, ds \\ &= -\int_e [\![\boldsymbol{u}]\!] \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{v}\}\!\}\boldsymbol{n}\, ds + \int_e \boldsymbol{g}^F \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{v}\}\!\}\boldsymbol{n}\, ds \end{aligned} \tag{69}$$

Dirichlet faces ($e \in \Gamma_D$): Here $\hat{\boldsymbol{u}} = \boldsymbol{g}^D$, one element K adjacent:

$$\int_e (\boldsymbol{g}^D - \boldsymbol{u}) \cdot (\nabla \boldsymbol{v} : \mathbb{C})\boldsymbol{n}\, ds = -\int_e \boldsymbol{u} \cdot (\mathbb{C} : \nabla \boldsymbol{v})\boldsymbol{n}\, ds + \int_e \boldsymbol{g}^D \cdot (\mathbb{C} : \nabla \boldsymbol{v})\boldsymbol{n}\, ds \tag{70}$$

Neumann faces ($e \in \Gamma_N$): $\hat{\boldsymbol{u}} = \boldsymbol{u}$, so $\hat{\boldsymbol{u}} - \boldsymbol{u} = \boldsymbol{0}$.

Assembly of Term II:

$$\begin{aligned} \text{Term II} = & \underbrace{-\sum_{e \in \Gamma_0 \cup \Gamma_F} \int_e [\![\boldsymbol{u}]\!] \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{v}\}\!\}\boldsymbol{n}\, ds - \sum_{e \in \Gamma_D} \int_e \boldsymbol{u} \cdot (\mathbb{C} : \nabla \boldsymbol{v})\boldsymbol{n}\, ds}_{\text{(II-a) depends on } \boldsymbol{u} \text{ and } \boldsymbol{v}} \\ & \underbrace{+ \sum_{e \in \Gamma_D} \int_e \boldsymbol{g}^D \cdot (\mathbb{C} : \nabla \boldsymbol{v})\boldsymbol{n}\, ds}_{\text{(II-b) Dirichlet data}} \ \underbrace{+ \sum_{e \in \Gamma_F} \int_e \boldsymbol{g}^F \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{v}\}\!\}\boldsymbol{n}\, ds}_{\text{(II-c) fault slip data}} \end{aligned} \tag{71}$$

Term III: Stress flux. *Interior faces ($e \in \Gamma_0$):* The stress flux is $\hat{\boldsymbol{\sigma}}\boldsymbol{n} = \{\!\{\mathbb{C} : \nabla \boldsymbol{u}\}\!\}\boldsymbol{n} + \eta_e \{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!])\}\!\}\boldsymbol{n}$. (We note that $\{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!])\}\!\} = \boldsymbol{r}_e([\![\boldsymbol{u}]\!])$ on e since $\boldsymbol{r}_e$ is a single polynomial on ω_e.) With the global minus from (58), the two-element assembly gives:

$$\begin{aligned} & -\int_e \boldsymbol{v}^- \cdot \hat{\boldsymbol{\sigma}}(+\boldsymbol{n})\, ds - \int_e \boldsymbol{v}^+ \cdot \hat{\boldsymbol{\sigma}}(-\boldsymbol{n})\, ds \\ &= -\int_e \boldsymbol{v}^- \cdot \hat{\boldsymbol{\sigma}}\boldsymbol{n}\, ds + \int_e \boldsymbol{v}^+ \cdot \hat{\boldsymbol{\sigma}}\boldsymbol{n}\, ds \\ &= -\int_e (\boldsymbol{v}^- - \boldsymbol{v}^+) \cdot \hat{\boldsymbol{\sigma}}\boldsymbol{n}\, ds = -\int_e [\![\boldsymbol{v}]\!] \cdot \hat{\boldsymbol{\sigma}}\boldsymbol{n}\, ds \end{aligned} \tag{72}$$

The global minus is absorbed, producing a *negative* coefficient on $[\![\boldsymbol{v}]\!]$. Expanding the stress flux:

$$= -\int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{u}\}\!\} \boldsymbol{n}\, ds - \eta_e \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!])\}\!\}\, ds \tag{73}$$

Fault faces ($e \in \Gamma_F$): Same assembly gives $-\int_e [\![\boldsymbol{v}]\!] \cdot \hat{\boldsymbol{\sigma}} \boldsymbol{n}\, ds$. Using linearity $\boldsymbol{r}_e([\![\boldsymbol{u}]\!] - \boldsymbol{g}^F) = \boldsymbol{r}_e([\![\boldsymbol{u}]\!]) - \boldsymbol{r}_e(\boldsymbol{g}^F)$:

$$\begin{aligned} &= -\int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{u}\}\!\} \boldsymbol{n}\, ds \\ &\quad - \eta_e \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!])\}\!\}\, ds + \eta_e \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e(\boldsymbol{g}^F)\}\!\}\, ds \end{aligned} \tag{74}$$

Dirichlet faces ($e \in \Gamma_D$): Single element, global minus not absorbed. Using $\boldsymbol{r}_e(\boldsymbol{u} - \boldsymbol{g}^D) = \boldsymbol{r}_e(\boldsymbol{u}) - \boldsymbol{r}_e(\boldsymbol{g}^D)$, and noting that $\boldsymbol{v} \cdot \boldsymbol{r}_e(\boldsymbol{u})\boldsymbol{n}$ denotes $v_i[\boldsymbol{r}_e(\boldsymbol{u})]_{ij} n_j$:

$$= -\int_e \boldsymbol{v} \cdot (\mathbb{C} : \nabla \boldsymbol{u}) \boldsymbol{n}\, ds - \eta_e \int_e \boldsymbol{v} \cdot \boldsymbol{r}_e(\boldsymbol{u}) \boldsymbol{n}\, ds + \eta_e \int_e \boldsymbol{v} \cdot \boldsymbol{r}_e(\boldsymbol{g}^D) \boldsymbol{n}\, ds \tag{75}$$

Neumann faces ($e \in \Gamma_N$): $\hat{\boldsymbol{\sigma}} \boldsymbol{n} = \boldsymbol{0}$, no contribution.

Assembly of Term III:

$$\begin{aligned} \text{Term III} = & \underbrace{-\sum_{e \in \Gamma_0 \cup \Gamma_F} \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{u}\}\!\} \boldsymbol{n}\, ds - \sum_{e \in \Gamma_D} \int_e \boldsymbol{v} \cdot (\mathbb{C} : \nabla \boldsymbol{u}) \boldsymbol{n}\, ds}_{\text{(III-a) symmetry}} \\ & \underbrace{- \sum_{e \in \Gamma_0 \cup \Gamma_F} \eta_e \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!])\}\!\}\, ds - \sum_{e \in \Gamma_D} \eta_e \int_e \boldsymbol{v} \cdot \boldsymbol{r}_e(\boldsymbol{u}) \boldsymbol{n}\, ds}_{\text{(III-b) stabilization (face form)}} \\ & \underbrace{+ \sum_{e \in \Gamma_F} \eta_e \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e(\boldsymbol{g}^F)\}\!\}\, ds}_{\text{(III-c) fault stab. data}} \underbrace{+ \sum_{e \in \Gamma_D} \eta_e \int_e \boldsymbol{v} \cdot \boldsymbol{r}_e(\boldsymbol{g}^D) \boldsymbol{n}\, ds}_{\text{(III-d) Dirichlet stab. data}} \end{aligned} \tag{76}$$

Term IV: Body force.

$$\text{(IV)} = \sum_{K \in \mathcal{T}_h} \int_K \boldsymbol{v} \cdot \boldsymbol{b}\, dx \tag{77}$$

Combining all terms. Summing (I) + (II) + (III) = (IV):

- **Bilinear form** $a(\boldsymbol{u}, \boldsymbol{v})$: terms (I), (II-a), (III-a), (III-b).
- **Linear form** $L(\boldsymbol{v})$: terms (IV), (II-b), (II-c), (III-c), (III-d), with appropriate signs.

Converting the stabilization to a volume integral.

Interior/fault stabilization (III-b), bilinear part. Setting $\boldsymbol{\varphi} = [\![\boldsymbol{v}]\!]$ in (60) and choosing $\boldsymbol{\tau} = \boldsymbol{r}_e([\![\boldsymbol{u}]\!]) \in \Sigma_h$:

$$\int_{\omega_e} \boldsymbol{r}_e([\![\boldsymbol{v}]\!]) \cdot \boldsymbol{r}_e([\![\boldsymbol{u}]\!])\, dx = -\int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!])\}\!\}\, ds \tag{78}$$

Therefore:

$$-\eta_e \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!])\}\!\}\, ds = +\eta_e \int_{\omega_e} \boldsymbol{r}_e([\![\boldsymbol{u}]\!]) \cdot \boldsymbol{r}_e([\![\boldsymbol{v}]\!])\, dx \tag{79}$$

The stabilization is *positive* in volume form.

Dirichlet stabilization (III-b), bilinear part. On a boundary face, the lifting reduces to $\int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{\varphi}) \cdot \boldsymbol{\tau}\, dx = -\int_e \boldsymbol{\varphi} \cdot \boldsymbol{\tau}\boldsymbol{n}\, ds$:

$$-\eta_e \int_e \boldsymbol{v} \cdot \boldsymbol{r}_e(\boldsymbol{u})\boldsymbol{n}\, ds = +\eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{v}) \cdot \boldsymbol{r}_e(\boldsymbol{u})\, dx \tag{80}$$

Also positive.

Fault data (III-c).

$$+\eta_e \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e(\boldsymbol{g}^F)\}\!\}\, ds = -\eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{g}^F) \cdot \boldsymbol{r}_e([\![\boldsymbol{v}]\!])\, dx \tag{81}$$

Enters $L(\boldsymbol{v})$ with sign flip: $+\eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{g}^F) \cdot \boldsymbol{r}_e([\![\boldsymbol{v}]\!])\, dx$.

Dirichlet data (III-d).

$$+\eta_e \int_e \boldsymbol{v} \cdot \boldsymbol{r}_e(\boldsymbol{g}^D)\boldsymbol{n}\, ds = -\eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{g}^D) \cdot \boldsymbol{r}_e(\boldsymbol{v})\, dx \tag{82}$$

Enters $L(\boldsymbol{v})$ with sign flip: $+\eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{g}^D) \cdot \boldsymbol{r}_e(\boldsymbol{v})\, dx$.

Summary of conversions.

$$\begin{aligned}
\text{(III-b) int./fault:} \quad & -\eta_e \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e([\![\boldsymbol{u}]\!])\}\!\}\, ds && = +\eta_e \int_{\omega_e} \boldsymbol{r}_e([\![\boldsymbol{u}]\!]) \cdot \boldsymbol{r}_e([\![\boldsymbol{v}]\!])\, dx \\
\text{(III-b) Dirichlet:} \quad & -\eta_e \int_e \boldsymbol{v} \cdot \boldsymbol{r}_e(\boldsymbol{u})\boldsymbol{n}\, ds && = +\eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{v}) \cdot \boldsymbol{r}_e(\boldsymbol{u})\, dx \\
\text{(III-c) fault data:} \quad & +\eta_e \int_e [\![\boldsymbol{v}]\!] \cdot \{\!\{\boldsymbol{r}_e(\boldsymbol{g}^F)\}\!\}\, ds && = -\eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{g}^F) \cdot \boldsymbol{r}_e([\![\boldsymbol{v}]\!])\, dx \\
\text{(III-d) Dirichlet data:} \quad & +\eta_e \int_e \boldsymbol{v} \cdot \boldsymbol{r}_e(\boldsymbol{g}^D)\boldsymbol{n}\, ds && = -\eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{g}^D) \cdot \boldsymbol{r}_e(\boldsymbol{v})\, dx
\end{aligned} \tag{83}$$

BR2 primal formulation. Find $\boldsymbol{u} \in V_h$ such that

$$a(\boldsymbol{u}, \boldsymbol{v}) = L(\boldsymbol{v}), \qquad \forall\, \boldsymbol{v} \in V_h \tag{84}$$

where the bilinear form is:

$$\begin{aligned}
a(\boldsymbol{u}, \boldsymbol{v}) = & \underbrace{\sum_{K \in \mathcal{T}_h} \int_K \nabla \boldsymbol{v} : \mathbb{C} : \nabla \boldsymbol{u}\, dx}_{\text{(I) volume / stiffness}} \\
& + \underbrace{\sum_{e \in \Gamma_0 \cup \Gamma_F} \eta_e \int_{\omega_e} \boldsymbol{r}_e([\![\boldsymbol{u}]\!]) \cdot \boldsymbol{r}_e([\![\boldsymbol{v}]\!])\, dx + \sum_{e \in \Gamma_D} \eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{u}) \cdot \boldsymbol{r}_e(\boldsymbol{v})\, dx}_{\text{(III-b) stabilization (lifting)}} \\
& - \underbrace{\sum_{e \in \Gamma_0 \cup \Gamma_F} \int_e \left([\![\boldsymbol{u}]\!] \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{v}\}\!\}\boldsymbol{n} + [\![\boldsymbol{v}]\!] \cdot \{\!\{\mathbb{C} : \nabla \boldsymbol{u}\}\!\}\boldsymbol{n}\right) ds}_{\text{(II-a) + (III-a) interior/fault: consistency + symmetry}} \\
& - \underbrace{\sum_{e \in \Gamma_D} \int_e \left(\boldsymbol{u} \cdot (\mathbb{C} : \nabla \boldsymbol{v})\boldsymbol{n} + \boldsymbol{v} \cdot (\mathbb{C} : \nabla \boldsymbol{u})\boldsymbol{n}\right) ds}_{\text{(II-a) + (III-a) Dirichlet: consistency + symmetry}}
\end{aligned} \tag{85}$$

and the linear form is:

$$
\begin{aligned}
L(\boldsymbol{v}) = & \underbrace{\sum_{K\in\mathcal{T}_h} \int_K \boldsymbol{v}\cdot\boldsymbol{b}\, dx}_{\text{from (IV) body force}} \\
& + \underbrace{\sum_{e\in\Gamma_D} \eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{g}^D)\cdot\boldsymbol{r}_e(\boldsymbol{v})\, dx}_{\text{from (III-d)}} - \underbrace{\sum_{e\in\Gamma_D} \int_e \boldsymbol{g}^D\cdot(\mathbb{C}:\nabla\boldsymbol{v})\boldsymbol{n}\, ds}_{\text{from (II-b)}} \\
& + \underbrace{\sum_{e\in\Gamma_F} \eta_e \int_{\omega_e} \boldsymbol{r}_e(\boldsymbol{g}^F)\cdot\boldsymbol{r}_e([\![\boldsymbol{v}]\!])\, dx}_{\text{from (III-c)}} - \underbrace{\sum_{e\in\Gamma_F} \int_e \boldsymbol{g}^F\cdot\{\!\{\mathbb{C}:\nabla\boldsymbol{v}\}\!\}\boldsymbol{n}\, ds}_{\text{from (II-c)}}
\end{aligned}
\tag{86}
$$

Structure of the bilinear form. With the convention $[\![\boldsymbol{w}]\!] = \boldsymbol{w}^- - \boldsymbol{w}^+$, the bilinear form (85) has the standard structure:

- The *volume term* (positive): standard continuous Galerkin stiffness.
- The *stabilization term* (positive): penalizes displacement jumps through the lifting operators, ensuring coercivity for η_e sufficiently large.
- The *consistency + symmetry terms* (negative): ensure that the exact solution satisfies the discrete equations and that $a(\cdot,\cdot)$ is symmetric.

This sign pattern—positive volume, positive stabilization, negative consistency/symmetry—is consistent with the standard SIPG/BR2 formulations in the literature (e.g., [**uphoff2023**]). The bilinear form is symmetric and, for η_e sufficiently large, coercive.

2.4.2 Interior Penalty (IP)

2.5 Algorithm

2.6 Verification: SCEC-SEAS-BP5

3 Dynamic Rupture 3D Half Space Problem

3.1 Strong Form

3.2 Weak Form

References

1. Arnold, D. N., Brezzi, F., Cockburn, B. & Marini, L. D. Unified analysis of discontinuous Galerkin methods for elliptic problems. *SIAM journal on numerical analysis,* **39** (2002).